

Seminar on Artificial Intelligence and Quantum Computing 2026

Friday, 16 October 2026

9:00 Day 2 Registration

9:30 Welcome Remarks - *Oscar Montiel Ross*

9:40 **A hybrid method for disease detection in cucumber plants, fruit and leaves based on vision transformers and convolutional neural networks.**

Kenia Picos, Ulises Orozco-Rosas.

Presenter: *Kenia Picos*; **Email:** kenia.picos@cetys.mx; **Institution:** *CETYS.*

10:00 **EMU: An Evolutionary Multi-Objective NAS Framework for Image Classification.**

Ricardo Rioda Santiago, Daniel E. Hernandez, Miguel Lopez-Montiel.

Presenter: *Ricardo Rioda Santiago*; **Email:** ricardo.santiago@itj.com; **Institution:** *ITJ.*

10:20 **Autonomous Navigation using Visual Memories: Integrating ROS, RGBD, and AI Algorithms.**

Kevin F. Rios-Alejo, Eder A. Rodríguez-Martínez, Ivan Y. Alba-Corpus, Moises J. Castro-Toscano, Julio C. Rodríguez-Quirón, Oleg Sergiyenko, Wendy Flores-Fuentes.

Presenter: *Kevin F. Rios-Alejo*; **Email:** flores.wendy@uabc.edu.mx; **Institution:** *UABC.*

10:40 **Scientometric analysis of application of quantum computing in healthcare.**

Oscar Eduardo Barreras Calderón, Juan Pablo García Vázquez, Daniel Cuevas-González, Juan Carlos Delgado Torres, Martín Aaron Sánchez Barajas, Eladio Altamira Colado, Kevin Daniel Plata Arellano, Edgar García Duarte.

Presenter: *Oscar Eduardo Barreras Calderón*; **Email:** barreras.oscar@uabc.edu.mx;

Institution: *UABC*

11:00 Keynote Lecture



Ethical Reasoning for Responsible Artificial Intelligence.

Dra. Lourdes Martínez Villaseñor

Email: lmartine@up.edu.mx

Institution: *Universidad Panamericana*

11:40 Coffee Break

12:00 Keynote Lecture



Robot Learning: A Reinforcement Learning Approach.

Dr. Ulises Orozco-Rosas

Email: ulises.orozco@cetys.mx

Institution: *CETYS Universidad*



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12:40 **Feasibility of Hybrid Classical-Quantum Architectures: An Exploratory Analysis of Partial Substitution in Flux.1 Generative Models.**

María de los Ángeles Cosío León, Joel Silos Sánchez, Geovani A. Monroy García, Anabel Martinez Vargas.

Presenter: María de los Ángeles Cosío León; **Email:** ma.cosio.leon@upp.edu.mx;

Institution: Universidad Politécnica de Pachuca.

13:00 **Hybrid Quantum Approaches for Colon Cancer Histopathology Image Classification.**

Kevin Daniel Plata Arellano, Juan Pablo García Vázquez, Jesús Eduardo Soto Vega.

Presenter: Kevin Daniel Plata Arellano; **Email:** pablo.garcia@uabc.edu.mx; **Institution:** UABC.

13:20 **Reconfigurable Neuromorphic Architectures on FPGA for Real-Time Processing.**

Teodoro Álvarez Sánchez, Jesus A. Álvarez Cedillo, Roberto Herrera Charles, Nashielly Yarzabal Coronel, Teresa Sarabia Alonso.

Presenter: Dr. Roberto Herrera Charles; **Email:** robcccharles@citedi.mx; **Institution:** CITEDI.

13:40 **Assessing PCA as a Baseline for Asteroid Spectral Classification: Robustness, Unsupervised Comparisons, and Quantum-Inspired Perspectives.**

Ana Victoria Ojeda Vera, Juan José Tapia Armenta, Luis Hector Sanchez Gamez.

Presenter: Ana Victoria Ojeda Vera; **Email:** aojeda@citedi.mx; **Institution:** CITEDI.

14:00 **Discriminating the Indistinguishable: Unsupervised Criteria for Texts in Highly Homogeneous Scientific Domains.**

Adán Hiraes Carbajal.

Presenter: Adán Hiraes Carbajal; **Email:** adan.hiraes@cetys.mx; **Institution:** CETYS Universidad.

14:20 **Noise-Aware Quantum AI Models Using Python and QuTiP for Decoherence, Gate Fidelity, and Quantum Feature Encoding.**

Rita Rosa Beltrán Castro, Oscar Montiel Ross, Moisés Sánchez Adame.

Presenter: Rita Rosa Beltrán Castro; **Email:** rbeltran@citedi.mx; **Institution:** CITEDI.

14:40 **Multi-Task Edge AI for Autonomous Security Patrolling: Joint Recognition of Anomalous Human Postures and Dangerous Objects on a Differential-Drive Robot.**

Leonardo D. Ortega-Lomeli, Luis C. Básaca-Preciado, Ulises Orozco-Rosas.

Presenter: Leonardo D. Ortega-Lomeli; **Email:** leonardo.ortega@cetys.mx;

Institution: CETYS Universidad.

15:00 **Day 2 Concludes**



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